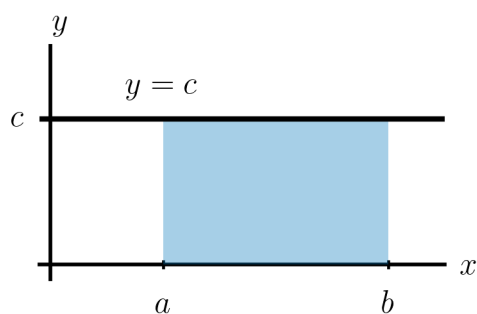


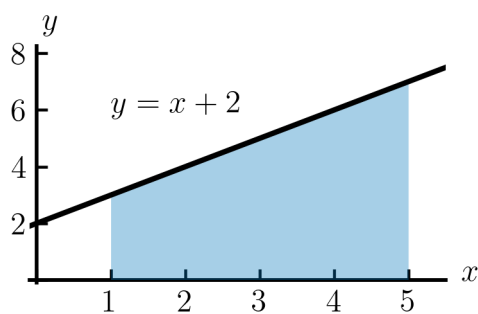
Evaluating Definite Integrals using Graphs and Geometry

In this lesson, we will use graphs and geometry to evaluate definite integrals.

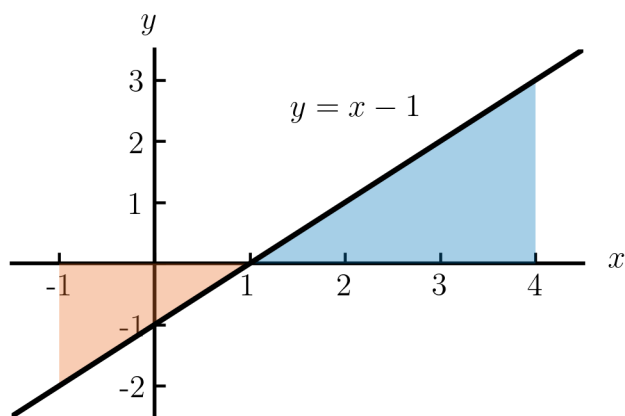
Say we want to evaluate $\int_a^b c \, dx$, where c is a constant. The definite integral gives the net area bounded by the graph of $y = c$ and the x -axis.



Example 1: Use geometry to evaluate $\int_1^5 (x + 2) \, dx$.

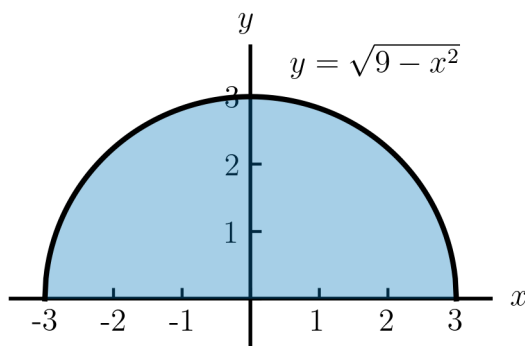


Example 2: Use geometry to evaluate $\int_{-1}^4 (x - 1) dx$.



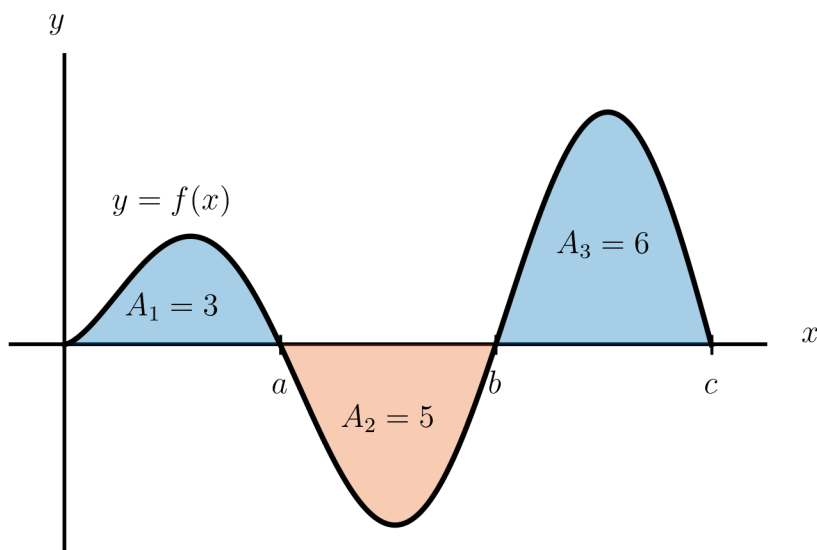
Now try these interactive problems on the website, to practice some new cases: an area under a curved function and using a graph with labeled areas to find a definite integral.

Interactive Problem 1: Use geometry to evaluate $\int_{-3}^3 \sqrt{9 - x^2} dx$.



- (a) On the graph, the definite integral $\int_{-3}^3 \sqrt{9 - x^2} dx$ is the
- (b) The formula for the area of a semicircle is
- (c) The radius of the semicircle is
- (d) Putting this all together, the definite integral is $\int_{-3}^3 \sqrt{9 - x^2} dx =$

Interactive Problem 2: The graph shows the areas $A_1 = 3$, $A_2 = 5$, and $A_3 = 6$ of regions bounded by the graph of $y = f(x)$ and the x -axis. We will use the graph to find $\int_0^b f(x) dx$ and $\int_a^c f(x) dx$.



(a) The areas involved in finding $\int_0^b f(x) dx$ are

(b) A_1 counts as

(c) A_2 counts as

(d) So, $\int_0^b f(x) dx =$

(e) The areas involved in finding $\int_a^c f(x) dx$ are

(f) The net area is $\int_a^c f(x) dx =$